Two-Dimensional Partition Theorem: Math Derivation

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Contents

Two-Dimensional Partition Theorem: Math Derivation

Source: planar specialization of Matoušek's Partition Theorem from *Efficient Partition Trees*. The cutting theorem used as a black box is the Chazelle-Friedman cutting machinery.

Goal

Input:

$$P = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$$

Choose:

$$2 \le s < n$$

Set:

$$r = \frac{n}{s}$$

Construct:

$$\Pi = \{(P_1, \Delta_1), \dots, (P_m, \Delta_m)\}$$

such that:

$$P_i \cap P_j = \varnothing \quad (i \neq j)$$

$$\bigcup_i P_i = P$$

$$P_i \subseteq \Delta_i$$

$$s \leq |P_i| < 2s$$

and:

$$\mathrm{cr}(\Pi) = O(\sqrt{r})$$

For a line h:

$$\mathrm{cr}_\Pi(h) = |\{i : h \text{ crosses } \Delta_i\}|$$

and:

$$\mathrm{cr}(\Pi) = \max_h \mathrm{cr}_\Pi(h)$$

Cutting Inputs

For n lines in the plane, a $(1/t)$-cutting has:

$$O(t^2)$$

cells, and each cell is crossed by:

$$O\left(\frac{n}{t}\right)$$

lines.

Weighted version:

For weighted lines (Q, w), define:

$$W = \sum_{q \in Q} w(q)$$

A weighted $(1/t)$-cutting satisfies for every cell C:

$$\sum_{q \in Q(C)} w(q) \leq \frac{W}{t}$$

where:

$$Q(C) = \{q \in Q : q \text{ crosses } C\}$$

In 2D, for some absolute constant A:

$$\#\text{faces} \le At^2$$

Step 1: Build the Finite Test Set Q

Use duality:

$$p = (a, b) \longleftrightarrow p^* : y = ax - b$$

$$\ell : y = ax - b \longleftrightarrow \ell^* = (a, b)$$

Incidence:

$$p \in \ell \iff \ell^* \in p^*$$

Above-below:

$$p \text{ above } \ell \iff \ell^* \text{ above } p^*$$

Verification:

Let:

$$p = (u, v)$$

and:

$$\ell : y = ax - b$$

Then:

$$p \text{ above } \ell \iff v > au - b$$

Also:

$$p^* : y = ux - v$$

and:

$$\ell^* = (a, b)$$

The dual point ℓ^* is above the dual line p^* iff:

$$b > ua - v$$

which is equivalent to:

$$v > au - b$$

Dualize P:

$$P^* = \{p_1^*, \dots, p_n^*\}$$

Choose:

$$t = \beta\sqrt{r}$$

with beta small enough that the cutting has at most r vertices.
Since:

$$O(t^2) = O(\beta^2 r) \leq r$$

the vertex set:

$$V = \{v_1, \dots, v_N\}$$

satisfies:

$$N \leq r$$

Dualize vertices back:

$$q_j = v_j^*$$

Define:

$$Q = \{q_1, \dots, q_N\}$$

Thus:

$$|Q| = N \leq r$$

Important object distinction:

$$R \subseteq P^*$$

is the sampled set of about t dual lines used to build the cutting.

$$V$$

is the set of cutting vertices, which are points in the dual plane.

$$Q = V^*$$

is a new set of primal lines.
Thus:

$$Q \not\subseteq P^*$$

The cutting parameter is:

$$t = \Theta(\sqrt{r})$$

so every dual cutting cell A is crossed by at most:

$$O\left(\frac{n}{\sqrt{r}}\right)$$

dual lines from P^*.

Step 2: Test Set Lemma

Assume Π is any simplicial partition with:

$$|P_i| \geq s$$

Define:

$$K_Q = \max_{q \in Q} \mathrm{cr}_\Pi(q)$$

Claim:

$$\forall h, \quad \mathrm{cr}_\Pi(h) \leq 3K_Q + O\left(\frac{n}{s\sqrt{r}}\right)$$

Proof skeleton:

Take any nonvertical line h. Its dual point is:

$$h^*$$

Let A be the triangular dual cutting cell with:

$$h^* \in \mathrm{int}(A)$$

Let:

$$A = \mathrm{conv}\{v_1, v_2, v_3\}$$

Define:

$$q_j = v_j^*$$

Then:

$$q_1, q_2, q_3 \in Q$$

Cells crossed by at least one of q_1, q_2, q_3 are at most:

$$3K_Q$$

Detailed count:

$$\text{cr}_\Pi(q_1) \leq K_Q$$

$$\text{cr}_\Pi(q_2) \leq K_Q$$

$$\text{cr}_\Pi(q_3) \leq K_Q$$

Therefore:

$$|\{\Delta_i : \Delta_i \text{ crossed by at least one of } q_1, q_2, q_3\}| \leq K_Q + K_Q + K_Q = 3K_Q$$

Call Δ_i bad if:

$$h \text{ crosses } \Delta_i$$

but:

$$q_1, q_2, q_3 \text{ do not cross } \Delta_i$$

Key geometric implication:

$$\Delta_i \text{ bad and } p \in P_i \implies p^* \cap A \neq \emptyset$$

Reason in sign language:

The three primal lines q_1, q_2, q_3 decompose the primal plane into sign cells.

The all-above cell is:

$$\{x : x \text{ is above } q_1, q_2, q_3\}$$

The all-below cell is:

$$\{x : x \text{ is below } q_1, q_2, q_3\}$$

If h crossed the all-above cell, there would be:

$$x \in h$$

with:

$$x \text{ above } q_1, q_2, q_3$$

By duality:

$$q_1^* = v_1, \quad q_2^* = v_2, \quad q_3^* = v_3$$

all lie on the same side of x^*.

Since:

$$x \in h$$

we have:

$$h^* \in x^*$$

So x^* is a line through an interior point of triangle A.

But a line through an interior point of a triangle cannot put all three triangle vertices on the same side. Contradiction.

The same applies to the all-below cell.

Therefore every sign cell crossed by h is mixed-sign.

Now let:

$$p \in P_i \subseteq \Delta_i$$

where Δ_i is bad.

Because q_1, q_2, q_3 do not cross Δ_i, the simplex Δ_i lies inside one sign cell of their arrangement.

Since h crosses Δ_i, that sign cell is crossed by h, hence is mixed-sign.

So p has mixed signs relative to:

$$q_1, q_2, q_3$$

By above-below duality, the vertices:

$$v_1, v_2, v_3$$

are not all on the same side of p^*.

If p^* did not cross A, then the convex triangle:

$$A = \text{conv}\{v_1, v_2, v_3\}$$

would lie entirely on one side of p^*, so all three vertices would be on the same side.

Contradiction. Thus:

$$p^* \cap A \neq \varnothing$$

Equivalently:

$$\text{bad primal simplex} \Longrightarrow \text{dual conflict with } A$$

Since A is a $(1/\sqrt{r})$-cutting cell:

$$|\{p \in P : p^* \cap A \neq \emptyset\}| = O\left(\frac{n}{\sqrt{r}}\right)$$

Bad groups are disjoint and each has at least s points, so:

$$\#\text{bad simplices} \leq O\left(\frac{n/\sqrt{r}}{s}\right)$$

Detailed division:

Let B be the number of bad simplices.

Each bad simplex contributes at least s original points:

$$|P_i| \geq s$$

The bad groups are disjoint, so total bad points are at least:

$$Bs$$

But the dual conflict bound gives total bad points at most:

$$O\left(\frac{n}{\sqrt{r}}\right)$$

Thus:

$$Bs \leq O\left(\frac{n}{\sqrt{r}}\right)$$

and:

$$B \leq O\left(\frac{n}{s\sqrt{r}}\right)$$

Thus:

$$\#\text{bad simplices} = O\left(\frac{n}{s\sqrt{r}}\right)$$

Therefore:

$$\text{cr}_\Pi(h) \leq 3K_Q + O\left(\frac{n}{s\sqrt{r}}\right)$$

Since:

$$r = \frac{n}{s}$$

we have:

$$\frac{n}{s\sqrt{r}} = \frac{r}{\sqrt{r}} = \sqrt{r}$$

So:

$$K_Q = O(\sqrt{r}) \Longrightarrow \mathrm{cr}_\Pi(h) = O(\sqrt{r})$$

Step 3: Construct Π for Fixed Q

At round i:

$$P^{(i)} = P \setminus (P_1 \cup \cdots \cup P_i)$$

$$n_i = |P^{(i)}|$$

For test line q:

$$\kappa_i(q) = |\{j \le i : q \text{ crosses } \Delta_j\}|$$

Define exponential weight:

$$w_i(q) = 2^{\kappa_i(q)}$$

Total weight:

$$W_i = \sum_{q \in Q} w_i(q)$$

Initially:

$$\kappa_0(q) = 0$$

$$w_0(q) = 1$$

$$W_0 = |Q|$$

If:

$$n_i < 2s$$

make the terminal group:

$$P_{i+1} = P^{(i)}$$

and stop.
Otherwise choose:

$$t_i = \alpha\sqrt{\frac{n_i}{s}}$$

where alpha is small enough that:

$$A\alpha^2 \leq 1$$

Then:

$$At_i^2 = A\alpha^2\frac{n_i}{s} \leq \frac{n_i}{s}$$

Known and chosen quantities in this formula:

$$n_i = |P^{(i)}|$$

is known at the start of round i.

$$s$$

is the fixed group-size parameter.

$$A$$

is the absolute constant from the weighted cutting theorem.

$$t_i$$

is chosen from these quantities.
The reason is the pigeonhole constraint:

$$\#\text{faces} \leq At_i^2 \leq \frac{n_i}{s}$$

so the average number of remaining points per face is at least:

$$\frac{n_i}{n_i/s} = s$$

Build a weighted $(1/t_i)$-cutting for (Q, w_i).
For every face C:

$$w_i(Q(C)) \leq \frac{W_i}{t_i}$$

Number of faces:

$$\leq \frac{n_i}{s}$$

Pigeonhole:

$$\exists F_{i+1} \quad |P^{(i)} \cap F_{i+1}| \geq s$$

Set:

$$\Delta_{i+1} = F_{i+1}$$

Choose:

$$P_{i+1} \subseteq P^{(i)} \cap \Delta_{i+1}$$

with:

$$|P_{i+1}| = s$$

Remove:

$$P^{(i+1)} = P^{(i)} \setminus P_{i+1}$$

Step 4: Weight Recurrence

Let:

$$Q_{i+1} = \{q \in Q \,:\, q \text{ crosses } \Delta_{i+1}\}$$

For $q \in Q_{i+1}$:

$$w_{i+1}(q) = 2w_i(q)$$

For $q \notin Q_{i+1}$:

$$w_{i+1}(q) = w_i(q)$$

Therefore:

$$W_{i+1} = W_i + w_i(Q_{i+1})$$

No-step derivation:

Split Q into crossing and noncrossing test lines:

$$Q = Q_{i+1} \cup (Q \setminus Q_{i+1})$$

Then:

$$W_{i+1} = \sum_{q \notin Q_{i+1}} w_{i+1}(q) + \sum_{q \in Q_{i+1}} w_{i+1}(q)$$

For noncrossing lines:

$$w_{i+1}(q) = w_i(q)$$

For crossing lines:

$$w_{i+1}(q) = 2w_i(q)$$

So:

$$W_{i+1} = \sum_{q \notin Q_{i+1}} w_i(q) + \sum_{q \in Q_{i+1}} 2w_i(q)$$

Rewrite:

$$W_{i+1} = \sum_{q \in Q} w_i(q) + \sum_{q \in Q_{i+1}} w_i(q)$$

Therefore:

$$W_{i+1} = W_i + w_i(Q_{i+1})$$

Since Δ_{i+1} is a cutting face:

$$w_i(Q_{i+1}) \le \frac{W_i}{t_i}$$

Hence:

$$W_{i+1} \le W_i \left(1 + \frac{1}{t_i}\right)$$

In nonterminal rounds:

$$n_i = n - is$$

and:

$$\frac{n_i}{s} = r - i$$

Thus:

$$t_i = \Theta(\sqrt{r - i})$$

so for a constant C:

$$\frac{1}{t_i} \le \frac{C}{\sqrt{r-i}}$$

Therefore:

$$W_{i+1} \le W_i \left(1 + \frac{C}{\sqrt{r-i}}\right)$$

The terminal round can at most double every weight:

$$W_m \le 2W_L$$

where L is the number of nonterminal rounds.

Step 5: Product Estimate

Since:

$$L \le r$$

we get:

$$W_m \le 2|Q| \prod_{i=0}^{L-1} \left(1 + \frac{C}{\sqrt{r-i}}\right)$$

Taking logs:

$$\log W_m \le O(1) + \log|Q| + \sum_{i=0}^{L-1} \log\left(1 + \frac{C}{\sqrt{r-i}}\right)$$

Using:

$$\log(1+x) \le x$$

gives:

$$\log W_m \le O(1) + \log|Q| + C\sum_{i=0}^{L-1} \frac{1}{\sqrt{r-i}}$$

Since:

$$\sum_{i=0}^{L-1} \frac{1}{\sqrt{r-i}} \le \sum_{j=1}^{r} \frac{1}{\sqrt{j}}$$

and:

$$\sum_{j=1}^{r} j^{-1/2} \leq 1 + \int_1^r x^{-1/2}\, dx$$

with:

$$\int_1^r x^{-1/2}\, dx = 2(\sqrt{r} - 1)$$

we obtain:

$$\sum_{j=1}^{r} j^{-1/2} = O(\sqrt{r})$$

Therefore:

$$\log W_m = O(\log |Q| + \sqrt{r})$$

Step 6: Bound K_Q

For any $q \in Q$:

$$w_m(q) = 2^{\kappa_m(q)}$$

and:

$$w_m(q) \leq W_m$$

Thus:

$$2^{\kappa_m(q)} \leq W_m$$

Taking logs:

$$\kappa_m(q) \leq \log_2 W_m$$

Hence:

$$\kappa_m(q) = O(\log |Q| + \sqrt{r})$$

Therefore:

$$K_Q = \max_{q \in Q} \kappa_m(q) = O(\log |Q| + \sqrt{r})$$

Since:

$$|Q| \leq r$$

we have:

$$\log |Q| \leq \log r$$

and:

$$\log r = O(\sqrt{r})$$

Thus:

$$K_Q = O(\sqrt{r})$$

Here:

$$\kappa_i(q)$$

is cumulative for one q up to round i.

$$K_Q$$

is the final maximum over all q in Q.
At the end:

$$K_Q = \max_{q \in Q} \kappa_m(q)$$

Step 7: Extend from Q to All Lines

From the Test Set Lemma:

$$\text{cr}_\Pi(h) \leq 3K_Q + O\left(\frac{n}{s\sqrt{r}}\right)$$

Substitute:

$$K_Q = O(\sqrt{r})$$

and:

$$\frac{n}{s\sqrt{r}} = \sqrt{r}$$

Then:

$$\text{cr}_\Pi(h) \leq 3O(\sqrt{r}) + O(\sqrt{r})$$

so:

$$\text{cr}_\Pi(h) = O(\sqrt{r})$$

for every line h.

Therefore:

$$\mathrm{cr}(\Pi) = O(\sqrt{r})$$

Final Statement

For every finite point set:

$$P \subset \mathbb{R}^2$$

and every:

$$2 \leq s < n$$

with:

$$r = \frac{n}{s}$$

there exists a simplicial partition:

$$\Pi = \{(P_1, \Delta_1), \ldots, (P_m, \Delta_m)\}$$

such that:

$$s \leq |P_i| < 2s$$

and:

$$\mathrm{cr}(\Pi) = O(\sqrt{r})$$

References

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